

17-1b Competition, Monopolies, and Cartels

Before considering the price and quantity of water that results from the duopoly of Jack and Jill, let's briefly discuss the outcomes that would result if the water market were either perfectly competitive or monopolistic. These two polar cases are natural benchmarks.

If the market for water were perfectly competitive, the production decisions of each firm would drive price to equal marginal cost. Because we have assumed that the marginal cost of pumping additional water is zero, the equilibrium price of water under perfect competition would be zero as well. The equilibrium quantity would then be **120** gallons. The price of water would reflect the cost of producing it, and the efficient quantity of water would be produced and consumed.

Now consider how a monopoly would behave. [Table 1](#) shows that total profit is maximized at a quantity of **60** gallons and a price of **\$60** a gallon. A profit-maximizing monopolist, therefore, would produce this quantity and charge this price. As is standard for monopolies, price would exceed marginal cost. The result would be inefficient because the quantity of water produced and consumed would fall short of the socially efficient level of **120** gallons.

What outcome should we expect from our duopolists? One possibility is that Jack and Jill get together and agree on the quantity of water to produce and the price to charge for it.

Such an agreement among firms over production and price is called

[**collusion** \(an agreement among firms in a market about quantities to produce or prices to charge\)](#)

, and the group of firms acting in unison is called a [**cartel** \(a group of firms acting in unison\)](#). Once a cartel is formed, the market is in effect served by a monopoly and we can apply our analysis from [Chapter 15](#). That is, if Jack and Jill collude, they will agree on the monopoly outcome because that outcome maximizes their total profit. Our two producers produce a total of **60** gallons, which sell at a price of **\$60** a gallon. Once again, price exceeds marginal cost, and the outcome is socially inefficient.

A cartel must agree not only on the total level of production but also on the amount produced by each member. In our case, Jack and Jill must agree on how to split the monopoly production of **60** gallons. Each member of the cartel will want a larger share of the market because a larger market share means larger profit. If Jack and Jill agree to split the market equally, each produces **30** gallons, the price is **\$60** a gallon, and each earns a profit of **\$1,800**.

